

Prathmesh Jain

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with experience building production-grade AI systems for BFSI workflows, specializing in agent orchestration, multi-agent workflows, retrieval-augmented generation (RAG), and scalable full-stack applications using FastAPI, React, and modern LLM frameworks.

EXPERIENCE

Tata Consultancy Services (TCS)

Oct 2024 – Present

Full Stack AI Engineer (Research & Innovation, CTO Team, BFSI Americas)

- **AI Compliance Screening Platform (FastAPI, React, LangChain, Playwright, MCP, AWS)**
- Built an end-to-end AI compliance screening platform automating KYC onboarding, sanctions screening, embargo verification, customer presence validation, association checks, and adverse media analysis, reducing investigation time from **1–2 days to under 5 minutes**.
- Engineered stateful agent workflows combining Google Search, Playwright-based web crawling, structured outputs, deterministic validation, fuzzy matching, and LLM reasoning to improve screening accuracy while minimizing hallucinations and false positives.
- Developed reusable FastMCP servers exposing search, crawling, and document-processing capabilities as standardized tools across multiple AI workflows, alongside asynchronous processing pipelines with Server-Sent Events for real-time workflow streaming.
- **GenAI Payment Repair System (FastAPI, LangChain, CrewAI, React)**
- Developed a multi-agent payment repair platform where repair and validation agents collaboratively analyzed, repaired, and validated MT/MX payment messages through iterative tool execution, recursive repair loops, and schema-driven validation.
- Reduced average repair time to **under 30 seconds**, enabling reviewers to focus primarily on validation while decreasing maker-reviewer feedback cycles by approximately **30%**; solution is expected to automate nearly **70%** of repair cases through straight-through processing (STP).
- Implemented workflow reliability using retries, fallbacks, persisted workflow state, human review checkpoints, rollback handling, and comprehensive Langfuse observability for tracing, latency analysis, evaluation, and token monitoring.
- **Commercial Lending AI / Credit Memo Generation (CrewAI, LangChain, FastAPI, React)**
- Engineered a multi-agent AI system with specialized agents for document parsing, financial spread analysis, and credit memo generation, automating lending workflows for large financial institutions.
- Built document intelligence pipelines to process 100+ page financial reports using parallel chunking, retrieval, structured extraction, and section-wise synthesis to populate standardized credit memo templates with minimal manual effort.
- Integrated tool-augmented reasoning using MCP-based web search, document retrieval, and structured outputs to improve financial analysis accuracy, while leveraging Langfuse for production tracing, evaluation, and workflow observability.

PERSONAL PROJECTS

BizAssist - AI Business Finance Assistant | *React, FastAPI, LangGraph, Qdrant, MongoDB*

[Live Demo](#) | [GitHub](#)

- Built a full-stack AI business assistant supporting document Q&A, uploaded file analysis, Google Sheets automation, and multi-step business workflows using **LangGraph**, React, FastAPI, MongoDB, and Qdrant.
- Designed a stateful multi-agent LangGraph workflow with chat, planner, executor, summarization, guardrails, interrupt-based clarification, and shared tool execution to efficiently handle both conversational and complex business tasks.
- Implemented a RAG pipeline using **Qdrant** for semantic retrieval over user documents, supporting knowledge-base search across PDFs, text files, invoices, reports, contracts, and image-based documents with user-scoped filtering.
- Built a semantic tool discovery system by indexing tool metadata in a persistent vector database, dynamically loading only relevant tools at runtime to reduce prompt context from ~3000 tokens to under 1000 tokens, improving scalability as new tools are added.
- Developed document ingestion pipelines with PDF extraction, multimodal OCR, and chunking, alongside real-time SSE streaming with live token generation, tool activity, planner/executor status, and Google Sheets integration.

Transcribe – Lecture Transcription Workspace | *React.js, FastAPI, Redis, PostgreSQL*

[Live Demo](#)

- Developed an AI-powered lecture transcription platform that converts long-form spiritual discourses into structured, searchable, and downloadable PDF transcripts.
- Built responsive frontend interfaces for authentication, file uploads, transcription history, live processing status, and document downloads.
- Implemented an asynchronous processing pipeline using Redis queues, FFmpeg-based media conversion, silence detection, and chunked processing for scalable long-form transcription.
- Optimized transcription quality for Sanskrit shlokas (verses) and multilingual pravachans (discourses) using Gemini, followed by AI-powered transcript aggregation into structured HTML and automated PDF generation for 50+ active users.

SmartKiranaBot – Telegram AI Operations Agent | *TypeScript, Node.js, Fastify, Vercel AI SDK, PostgreSQL*

[Live Demo](#) | [GitHub](#)

- Engineered an end-to-end, Telegram-first AI agent for retail operations, executing Zod-validated tool calls across inventory, draft billing, GST line calculations, customer khata, and analytics.
- Enforced strict backend guardrails with atomic non-negative stock updates in Drizzle/PostgreSQL and conditional service locks to prevent overselling and inventory race conditions.
- Implemented multi-turn bill editing, webhook update idempotency, and durable shop memory persisting cross-session settings across conversation resets.
- Built artifact generation tools to dynamically render and deliver real GST invoice PDFs and analytical PowerPoint decks directly within the Telegram chat pipeline.

TECHNICAL SKILLS

AI / LLM: LangChain, LangGraph, CrewAI, RAG, Vector Search, Tool Calling, Multi-Agent Systems

Languages: Python, TypeScript, JavaScript

Backend: FastAPI, Node.js

Frontend: React.js, Next.js, Tailwind CSS

Databases: MongoDB, PostgreSQL, Qdrant

Tools & APIs: Docker, Redis, Git, Postman, Playwright

EDUCATION

Vidyalankar Institute of Technology

Bachelors in Computer Engineering

GPA: 9.2 / 10

Mumbai, IN

May 2024